Enhanced optical chirality with directional emission of Surface Plasmon Polaritons for chiral sensing applications

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Supplementary material

1. Mie modes of the structure

As discussed at the beginning of the results section, the extinction resonances of the unit cell with CPL illumination can be seen as a mixed contribution from resonances with linearly polarized illumination in the two axes x and y. Here, in Fig. S1, the scattering (Fig. S1a) and absorption (Fig. S1b) cross sections of the example unit cell (shown in Fig. 2 in the main text) upon x and y −linearly polarized and RCP and LCP illuminations are shown. It can be observed that for all illuminations, scattering dominates over absorption.

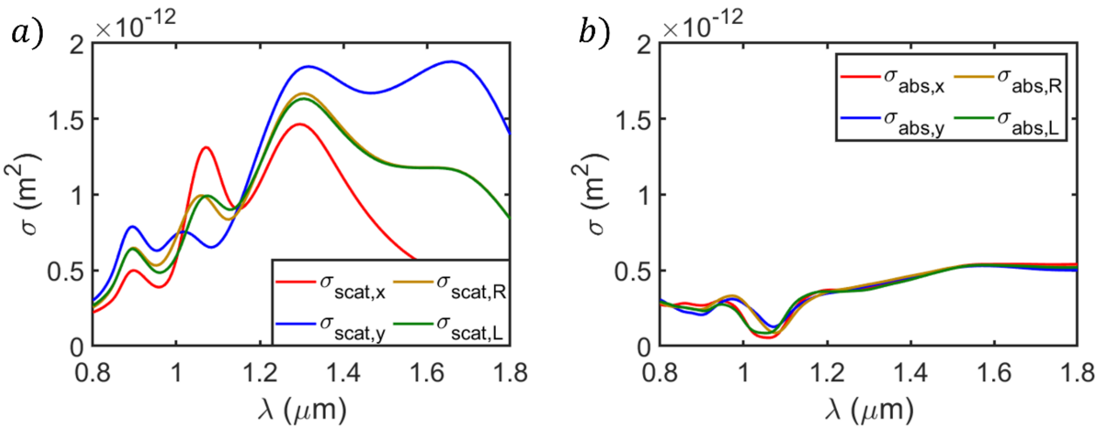

Fig. S1. a) Scattering cross section of the example geometry. b) Absorption cross section of the example geometry. The extinction cross section of this example geometry, sum of the scattering and the absorption, is depicted in Fig. 2 in the main text.

2. Extinction cross section of the geometrical parameters study

As has been already established, to reach directional plasmon chiral sensing a wavelength match between the CD signal peak, the extinction dipolar resonances and the grating working wavelength is necessary. Fig. S2 shows the extinction cross-sections for the analysed cases in Fig. 4. It can be seen the merging or splitting between x and y modes, depending on how the unit cell geometry changes.

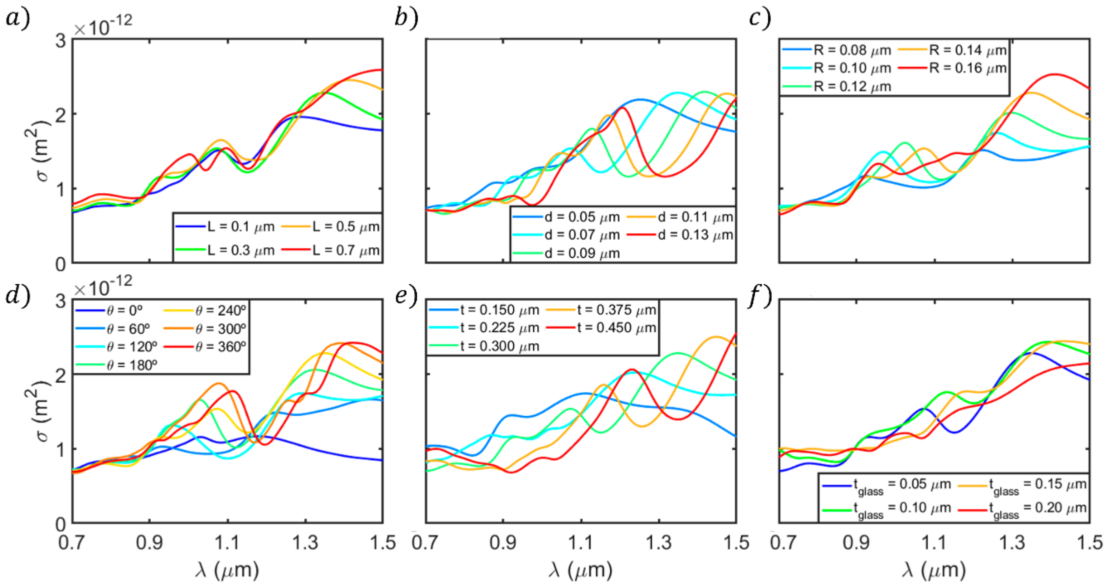

Fig. S2. Extinction cross-section upon variation of geometrical parameters in the unit cell structure. a) Central length L. b) Width d. c) Curve radius R. d) Angle θ. e) Height t. f) Glass layer thickness t_{glass}.

By comparison between Fig. 4 and Fig. S2, it can be observed that a match between the CD signal peak and the main dipole resonances only occurs for few cases. A strong manipulation of the CD signal can be achieved when varying the closing angle θ. In particular, the $\theta = 180°$ and $\theta = 300°$ cases display this match. However, the later case leads to several computational and fabrication issues and challenges, making the former a more practical choice for a proof-of-concept device, at the price of a smaller CD signal.

3. Study of directional plasmonic efficiency

The conclusion that directional emission requires the metasurface excitation to be at wavelengths compatible with the major unit cell dipolar resonances and diffraction grating modes is supported by the directional plasmonic efficiency shown in Fig. S3. Although strong plasmonic modes are found at many different wavelength positions, most of them are non-directional, with efficiencies around 60-70% of the total power.

However, the $L = 300$ nm case in Fig. S3a shows a remarkable 92% peak at around $\lambda = 1150$ nm. Here, the wavelengths match and directional plasmon is achieved. If the unit cell length is increased, the resonance spectral position redshifts, where other diffraction grating modes are located, thus, the directional plasmon vanishes. However, for similar lengths, if the resonances are blue-shifted (for example, by reducing the curvature angle θ) the wavelength match occurs again and the directional effect is recovered, as can be seen for θ = 120-180° in Fig. S3b. Fig. S3c shows how the grating effect is lost upon increase of the glass layer depth, leading to blurry modes for depths over 100 nm.

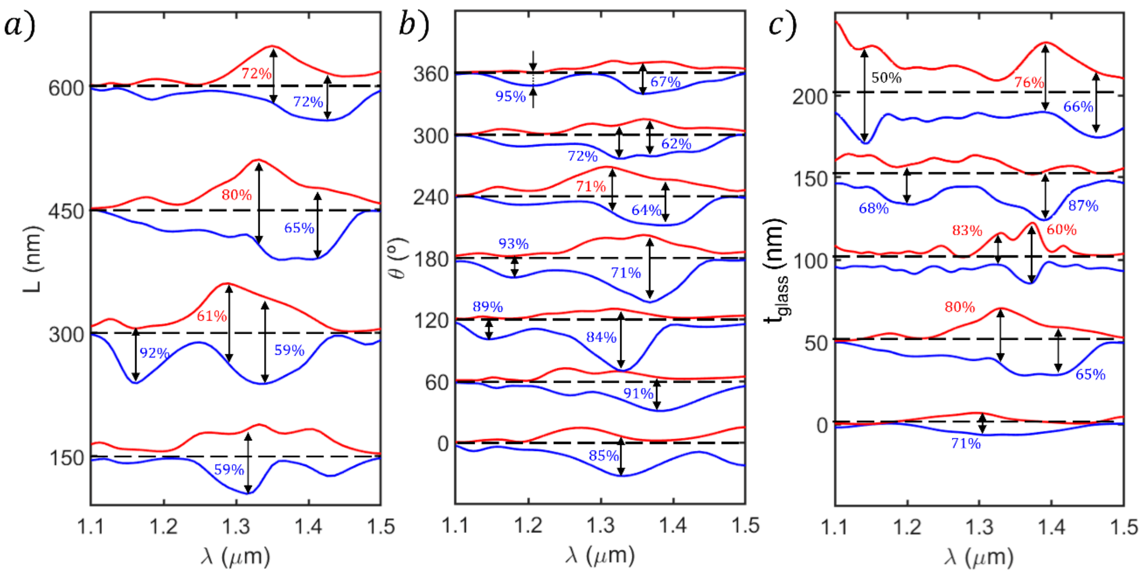

Fig. S3. Plasmon transmittance measured at 4.5 μm left and right from the center of the metasurface upon variation of several geometrical parameters with LCP illumination. The metasurface period was set to $\Lambda = 1200$ nm and the geometrical parameters, unless changed, were $(L, d, R, t, t_{glass}) = (500, 70, 140, 300, 50)$ nm with $\theta = 270°$.

4. Absorption and scattering cross sections of the optimal configuration

Fig. S4 shows the absorption and scattering cross sections of the chosen configuration in Fig. 5. As it happened with the example structure, scattering dominates over absorption and the resonances found here mirror the combination of dipolar modes for linearly polarized light.

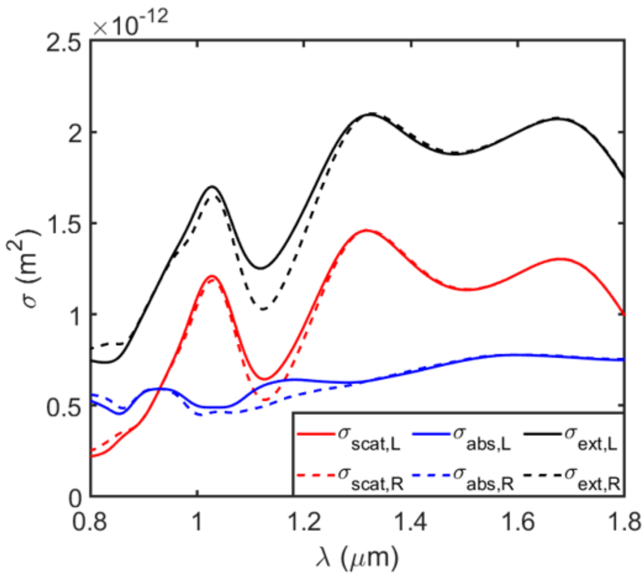

Fig. S4. Absorption, scattering and extinction cross-sections for LCP and RCP illuminations in the chosen configuration in Fig. 5. The normalized difference between the extinction cross-sections yields the CD signal shown in Fig. 5a.